

Technical Architecture and Technology Stack

Project: AI Travel Assistant for Accessible Aviation: Hotel, Booking Services & FlightRadar Integration

Client: Your Accessible Flight

Team: Team 36 – AI Travel Assistant for Accessible Aviation Team B

Date: 20/08/2026

Source: Client discussions – 07/08/2026 and subsequent client discussion

Proposed architecture

The AI travel assistant will use a client-server architecture consisting of a frontend application, Node.js backend service, Firestore database, an AI service, and external travel information providers.

The frontend application provides a user interface where users can enter travel requirements, accessibility preferences, and interact with the AI assistant. User requests are sent to the Node.js backend, which is the central processing layer.

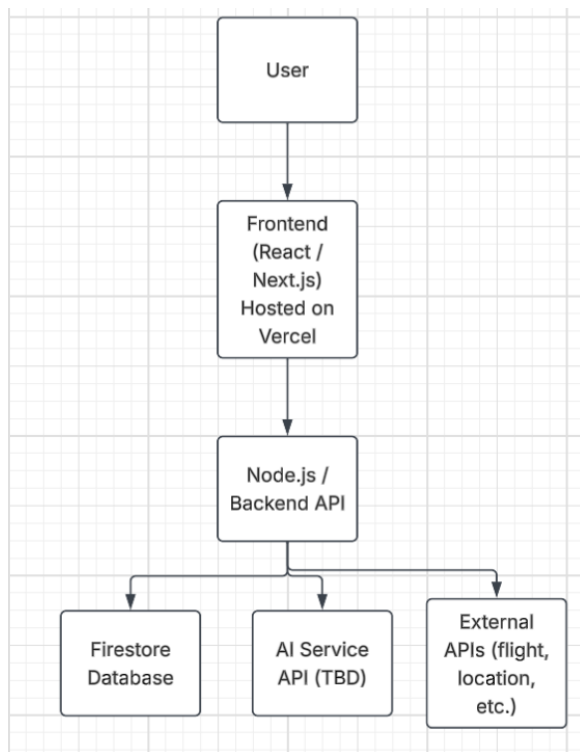
The backend communicates with the AI service to retrieve travel-related information from external APIs, and storing user information in Firestore. Firestore will be used to store user profiles, accessibility preferences, and conversation history to support personalised recommendations.

The AI service integration layer will allow the system to use an existing AI provider through an API rather than developing a custom AI model. The AI service will provide a natural language processing capabilities required in order to under the user requests identifying the accessibility requirements, and to generate relevant travel recommendations.

The system will be deployed using Vercel, which will host the frontend application and provide the deployment environment for the web application. The deployed frontend communicates with the backend service through http requests.

This proposed architecture supports the client's requirement of providing a single interface that combines accessibility information from multiple travel providers while having existing airline, accommodation, and booking platforms.

Architecture Diagram



AI service integration

The ai travel assistant will integrate with an existing AI service provider through an API rather than developing and training a custom AI model.

This approach follows the client's requirements of using an existing AI service provider/APIs

The AI service will provide the conversational intelligence required by the Travel Assistant, including:

- To understand the natural language travel requests
- To identify user accessibility requirements
- Processing user preferences and constraints
- Generate personalised travel recommendations

When selecting the specific AI service provider. The AI service provider will be evaluated and selected during the development process. Selection criteria will include:

- Quality of AI-generated responses
- API availability and reliability.
- Cost considerations.
- Privacy and data handling requirements
- Suitability for accessible-focused travel assistance

Database integration

The system will use google Firestore as the primary database.

Firestore is used because it provides a scalable cloud-based database solution that stores application data required to provide personal assistance.

The database will store:

- User profiles (email and password)
- Accessibility preferences
- Saved travel preferences
- Conversation history
- User interactions with the travel assistant

The node.js backend will communicate with Firestore to securely retrieve and update user information.

External Services and APIs

The system will investigate and integrate external services that can provide relevant travel information

Potential external services could include:

Service	Purpose
AI Service Provider API	Provides a AI to have a conversation with and have natural language processing capabilities
Travel information APIs	Provides flight, accommodation, travel-related information
Flight information Services	Provides flight availability and status information.
Maps API	Provides the location, navigation and distance information to the location entered
Weather API	Provides destination weather information where required

these selection of external API are subject to change

Technology Stack

Component	Technology
Frontend application	React / Next.js
Backend Service	Node.js
Database	Google Firestore
Deployment Platform	Vercel
AI integration	Existing AI Service Provider(OpenAI, Gemini, still to be picked)
External Services	Travel, Maps, Weather, and Accessibility APIs

Technical Dependencies

The technical dependencies required for the system include:

- Frontend application framework (React / Next.js)
- Node.js backend environment
- Firestore database service
- Vercel deployment platform
- AI service provider API
- Stable internet connection
- Authentication and security measures if needed

Risks and mitigation

Risks	Mitigation
AI service becomes unavailable	Implement error handling and provide users with appropriate messages to warn
AI-generated information is inaccurate	Provide a message for information may be inaccurate and validate the response by adding external travel information sources where possible
External API failures	Handle API errors

Firestore service interruption	Add database error handling and backup data to prevent loss of important user data
API usage limits	Monitor API usage
Accessibility information is incomplete	Communicate only the available information and avoid making assumptions